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Mental health outcomes and socio-demographics as determinants of students' academic performance and smartphone addiction at the University of Cape Coast

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Abstract

Objectives

Smartphone addiction is a significant mental health concern that affects students' academic performance, with attendant consequences on their psychological well-being, and yet only a handful of empirical research has been conducted on this area in Sub-Saharan Africa. The study, therefore, assessed mental health outcomes and socio-demographics as determinants of students' academic performance and smartphone addiction among students.

Methods

Employing a descriptive and analytical cross-sectional study design, purposive sampling was used to recruit 227 School of Nursing and Midwifery students. The study variables were measured with the Depression Anxiety Stress Scale (DASS-21) and Smartphone Addiction Scale.

Results

The results showed that 89 (39.2%) male and 138 (60.8%) female nursing students participated in the study. Employing descriptive statistics, we found a 43.6% prevalence of smartphone addiction. Pearson correlations showed significant associations between mental health outcomes and smartphone addiction. Utilising multi-linear regressions, socio-demographic factors and mental health outcomes did not jointly predict smartphone addiction, whereas mental health outcomes independently predicted smartphone addiction. Hierarchical regressions revealed that student's socio-demographics and mental health outcomes significantly influenced their academic performance, particularly for variables such as 'gender', 'age', 'level of study' and 'mental health outcomes'. Multivariate Analysis of Variance (MANOVA) revealed that age, level of study, marital status and mental health outcomes reported main and interaction effects on academic performance and smartphone addiction.

Conclusion

The study observed that smartphone addiction is remarkably high among students and should be addressed, while concerted efforts should be made to address depression, anxiety and stress among students, which pushes them into addictive behaviours. Recommendations for clinicians, educationists, social media experts and information technology professionals were drawn for positive coping, adjustment, and well-being.

Background

Smartphones have become a rapidly growing technology industry segment [41]. It is estimated that about 2.8 billion people globally possess smartphones [51,52]. Smartphone applications can be used in various settings, including social activities, education, and health management. However, over-dependence on smartphones may lead to addiction (chemical or process), resulting in maladaptive behaviours [2,16,26]. Evidence suggests associations between Smartphone Addiction (SA) and mental health issues such as depression and anxiety [7,42,33], poor sleep and poor quality of life [49]. In a longitudinal study conducted 2023, the researchers evaluated the association between religious actions, depressive symptoms, and stress among 413 students of nursing education and found a high prevalence of depressive symptoms and anxiety. Depression was more common among students in the first two years of studies and female medical students [43].

It is argued that tertiary education students are the fastest adopters of smartphone technology [19,3]. A study among students reported females to be significantly more users of smartphones than males [44]. In a more recent study conducted in 2024 which explored nomophobia and the extent and potential predictors among 636 university students; the results showed that nomophobia (overuse of mobile phones and the fear of missing out) varied in severity from mild to severe. Females experienced nomophobia more frequently than males. They discovered that gender, phone-checking frequency, number of calls made and received per day, and number of texts received in a single day predicted nomophobia [4].

Similarly, a survey among university students found that excessive video games were associated with a low Cumulative Grade Point Average (CGPA) [45]. Another study found a negative association between text messaging and academic performance [38]. However, only some studies investigated smartphone usage patterns in Ghana. Another study explored the utilisation of smartphones in teaching and learning in selected universities in Ghana [49]. Furthermore, another study was conducted in 2017 which investigated the use of smartphone applications for clinical decision-making in Ghana; however, little is known about smartphone addiction, its implications on mental health and academic performance among university students in Ghana and there is no known study in the University of Cape Coast [47].

There is increasing research on the obsessive attachment of people to the internet and smartphone use and the consequent physical, psychological, and social health implications, but studies are limited in Sub-Sahara Africa. Besides the social issues, such as isolation from immediate social interaction associated with over-dependence on the internet and mobile phones, recent findings revealed the potential for long-term health risks associated with these addictive smartphone usage habits [30]. More so, the COVID-19 pandemic may continue to have far-reaching consequences, particularly on our socioeconomic and educational lives. Another study examined the roles of a drug use disorder, neurotic health symptoms, and pathological smartphone use in COVID-19 lockdown-related stress. Results of the study revealed that COVID-19 lockdown stress was a result of the joint function of a drug use disorder, neurotic health symptoms, and pathological smartphone use. In particular, during the peak of COVID-19, almost all organisations, including academic institutions, resorted to online-based learning, thereby increasing the adoption of smartphones to learn and transact activities [39]. The University of Cape

Coast is not an exception. These activities may worsen smartphone dependence, significantly affecting psychosocial health and academic performance. This study is particularly relevant because although the pandemic seems to be under control, addicted students may struggle and require support for psychosocial life and academic performance. There is, however, scarce literature on the African population [40]. Findings from this survey will enable university management to plan and implement academic and counselling interventions for students to allow them to complete their education successfully. Furthermore, Ghana is a signatory to the 2030 Agenda for Sustainable Development; the outcome and impact of this project will enable the country to progress in milestones concerning health, education, socioeconomic growth, social inclusion, and protection. Specifically, the project will contribute to the achievement of the UN Sustainable Development Goals (SDG) by ensuring healthy lives and promoting the well-being of all at all ages (SDG 3), Ensuring inclusive and equitable quality education and promoting lifelong learning opportunities for all (SDG 4).

The interwoven nature of mental health, demographics, smartphone addiction and academic performance in the life of an average student makes it challenging for them to discontinue problematic behaviours which are at variance with their psychological well-being. Therefore, it is pertinent to investigate smartphone addiction, mental health outcomes and the impact on students' academic performance.

Aim of the study

This study investigated the mental health outcomes, smartphone addiction, and the impact on students' academic performance.

The following research hypotheses were investigated to achieve the study's aim.

Research Hypotheses

1. The prevalence of smartphone addiction will be high among students
2. There will be a significant relationship between smartphone usage and depression, anxiety, and stress among students.
3. Socio-demographic factors and mental health outcomes will jointly predict smartphone addiction among students.
4. Students' socio-demographic factors and mental health outcomes will significantly influence their academic performance.
5. Students' socio-demographic factors and mental health outcomes will significantly influence their academic performance and smartphone addiction.

METHODS

Research design

The study adopted both a descriptive and analytical cross-sectional study design. This study aimed to survey the students regarding smartphone addiction and its impact on mental health and academic performance at one point.

Population

Regular students at the University of Cape Coast School of Nursing and Midwifery represented the study's target population. The accessible population were all regular students on campus at the time of the survey.

Sample and sampling procedure

The study adopted a total sampling strategy, a set of purposive sampling methods that target the entire population with a particular set of characteristics. For example, the population of regular students in the School of Nursing and Midwifery is estimated at five hundred (500); only students who consented to participate were included. However, the authors administered 300 copies of the instruments, and only 227 copies were usable for the final analyses, representing a response rate of 75.67 per cent. The reason for purposively sampling the opinions of this particular population was further justified by one study which revealed that getting to a population is almost always done purposively since the researcher knows the population he is interested in and goes directly to them for such information [10]. Some other studies further justified the choice of adopting this sampling strategy when they emphasised that purposive sampling helps the researcher to concentrate on people with particular characteristics (which in this case is smartphone addiction and challenging academic performance) who will better be able to assist in providing relevant data and information on the subject matter [17,48]. In terms of the sample size, a renowned scholars reviewed the contemporary approach to calculating sample size with finite and non-finite populations and provided two empirical options: the n-hat method introduced in 2013 and the Multistage Non-finite Population (MNP) or n-omega method – which in either case, the minimum sample size ranges from 30 to 40 participants/counts [31]. On the other hand, another scholar propounded that sample size can be justified through any of these six approaches: i) collecting data from the entire population in question, (ii) obtaining a sample size based on resource challenges, (iii) carrying out a-priori power analysis, (iv) planning for a desired precision, (v) using heuristics, or (vi) out rightly acknowledging the absence of a justification[24]. In addition, after an extensive review, recent studies suggested that a sample between 160 and 300 valid observations is well suited for multivariate analysis [34]. Given this, the sample size justification in this study is pegged on the triangular standpoint of Memon et al. studies, who suggested between 160 and 300 samples is sufficient, coupled with Louangrath study's findings as well as Laken's' second approach in justifying sample size in a scientific study [24,31,34].

Study setting/location

The study setting was the University of Cape Coast, School of Nursing and Midwifery, comprising all regular undergraduate and postgraduate students. The University of Cape Coast is the largest public

university in Ghana, ranked number one in Ghana and West Africa [50]. The School of Nursing and Midwifery is one of the four schools in the College of Health and Allied Sciences. The school comprises four Departments: Adult Health, Mental Health, Public Health and Maternal and Child Health.

Data collection and procedure

Five research assistants were recruited to help with the data collection when the Institutional Review Board, University of Cape Coast, approved the study. All research assistants hold at least a Bachelor of Science (BSc) degree. The Department of Mental Health provided training sessions to the research assistants before the commencement of the data collection. The training included communication skills, confidentiality, and data protection. Written consent was obtained from participants before the commencing of the research. Also, none of the participants were minors.

Structured instruments were used to collect data comprising socio-demographic information, mental health outcomes such as depression, anxiety and stress and smartphone addiction data, as described briefly below.

Measures

Mental health outcomes. Mental health outcomes were assessed with the valid and reliable Depression Anxiety Stress Scale (DASS-21), [8,32]. Respondents were asked how often they experienced the mental states described in the items (e.g., “I find it difficult to work up the initiative to do things”) on a 4-point Likert-type scale (0 = never, 3 = almost always). Higher scores indicated high emotional distress. The Cronbach’s alpha coefficients of depression, anxiety, and stress are 0.90, 0.82, and 0.93, respectively. However, in the main study, the Cronbach’s alpha coefficient for DASS was 0.96, and the norm is 23.06.

Smartphone Addiction. Smartphone addiction refers to obsessive and compulsive reliance on smartphone use to the extent that physiological, mental, or emotional outcomes occur. Survey items assessed tolerance and withdrawal symptoms (e.g., “I am always thinking that I have a message on my smartphone”). Respondents expressed feelings about using their smartphones on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Higher scores represented high smartphone addiction (Cho et al., 2012). In addition, students were also provided with ranges to indicate their Cumulative Grade Point Average (CGPA). Also, the Cronbach’s alpha for Smartphone Addiction Scale is 0.81, and the norm is 52.30. As a result, the reliability test of the two (2) scales utilised in this study showed high reliability with acceptable internal consistency.

Data analyses

The data was processed and analysed. The psychometric properties of the constructs and tests of the proposed model and hypotheses were evaluated utilising IBM SPSS Statistics version 23. A descriptive analysis of the prevalence of smartphone addiction was presented. Frequencies, percentages, correlations and mean values were obtained. Also, inferential statistics, such as multiple regressions, hierarchical regression, and multivariate analysis of variance (MANOVA), were employed to determine

how participants' mental health outcomes and socio-demographics predicted smartphone addiction and students' academic performance.

Ethics and data management

Information was recorded to ensure anonymity and privacy protection of data so that participants could not be identified directly or indirectly through identifiers. Principles enshrined in the Belmont Report were adopted, and field assistants received training on the importance of adhering to the principles of anonymity and privacy to ensure that the research was conducted responsibly and ethically and that the rights and welfare of participants were protected. Participants' data were coded. The data was backed up remotely. All electronic communication was pass-worded or biometrically accessed. Hard and soft copies of collected data will be destroyed after five years. Before participating in this study, the purpose of the study was explained to the potential participants in an easy-to-understand language devoid of jargon. Adequate opportunity was given to participants to consider all options and all questions regarding the study. All participants provided informed consent for their voluntary participation. All procedures were performed by the 1964 Helsinki Declaration and comparable ethical standards. Ethical approval was obtained from the Institutional Review Board, University of Cape Coast.

Results

Descriptive results in Table 1 indicated that two hundred and twenty-seven (227) students were assessed, with ages ranging from 18 to 21 years and 22 to 25 years being more in number (170; 74.89%) compared to those above 30 years who were 37 (16.3%); those between 26 to 30 years who were 18 (7.9%) and students below 18 years who were 2 (0.9%). Participants' gender showed females were more in number, 138 (60.8%) compared to males, 29 (12.8%), while 9 (4.0%) preferred not to disclose. When participants were quizzed about their religious leaning, 209 (92.1%) responded that they were Christians, while 16 (7.0%) were of the Islam religion and 2 (0.9%). The stage of study or levels revealed that 64 (28.2%) were in level 100 of study; 27 (11.9%) were in level 200; 25 (11.0%) reported that they were in 300 level of study; 99 (43.6%) said that they were in 400 level of study; 9 (3.9%) responded that they were in 500 level or having an extra year in school, while 3 (1.3%) responded that they were in post-graduate study level.

Table 1: Descriptive summary of demographic variables among participants (*n* = 227)

Demographic Variables	*n*	%
Gender		
Male	89	39.2
Female	138	60.8
Marital Status		
Single	189	83.3
Married	29	12.8
Prefer not to disclose	9	4.0
Religion		
Christianity	209	92.1
Islam	16	7.0
Traditional	2	0.9
Level of Study		
100L	64	28.2
200L	27	11.9
300L	25	11.0
400L	99	43.6
500L	9	3.9
Post-graduate	3	1.3
Age		
Below 18 years	2	0.9
18-21 years	84	37.0
22-25 years	86	37.9
26-30 years	18	7.9
Above 30 years	37	16.3
Academic Performance (CGPA)		
Less than 1.0	1	0.4
1.0 to 1.9	10	4.4
2.0 to 2.4	13	5.7

2.5 to 2.9	68	30.0
3.0 to 3.5	93	41.0
3.6 to 4.0	42	18.5
Smart Phone Addiction (Prevalence)		
High Level	99	43.6
Low Level	128	56.4
Rate of Smart-phone usage		
Drop	64	28.2
Rise	163	71.8

The academic performance of participants was also assessed based on the Cumulative Grade Point Average (CGPA) and descriptive results revealed that 1 (0.4%) had less than 1.0 CGPA; where 10 (4.4%) had 1.0 to 1.9 CGPA, 13 (5.7%) reported a CGPA of 2.0 to 2.4; 68 (30.0%) reported a CGPA score of 2.5 to 2.9; majority 93 (41.0%) had had a CGPA score of 3.0 to 3.5; while 42 (18.5%) reported a CGPA score of 3.6 to 4.0. Participants were analysed on the rate at which their smartphone use led to the drop and rise of their CGPA score; 64 (28.2%) reported a drop in their CGPA score, while 163 (71.8%) reported an increase in their CGPA score. Lastly, students' smartphone addiction level was descriptively analysed, and the outcome indicated a high level of 43.6% and a low level of 56. 4%, this amounts to a smartphone prevalence rate of 43.6 per cent. Based on this, results from Table 1 showed that the first hypothesis, which stated that the prevalence of smartphone addiction would be high among students, was confirmed.

Table 2: Summary of means, standard deviations and correlation matrix of study variables in exploring the relationship between mental health outcomes (depression, anxiety and stress) and Smartphone Addiction (SMA) among the students

Variables	SMA	Depression	Anxiety	Stress	Mean	SD
SMA	1				52.30	12.40
Depression	0.16*	1			7.59	6.68
Anxiety	0.20**	0.89**	1		7.26	6.14
Stress	0.15*	0.92**	0.89**	1	8.21	7.09

*Correlation is significant at the 0.01 level and at the 0.05 level. Note: SMA = Smart-phone Addiction

A correlation analysis was conducted using the Pearson Product Moment Correlation Coefficient to investigate the magnitude in which all mental health outcomes were connected with smartphone addiction, as presented in Table 2 above. Firstly, the depression component of mental health outcomes is directly related to smartphone addiction (r = 0.16, p<0.05), meaning that a high level of depression necessitates smartphone addiction. Secondly, the anxiety component is directly related to smartphone addiction and depression (r = 0.20, p<0.01; r = 0.89, p<0.01), respectively. This implies that a high level of anxiety brings about smartphone addiction and compounds the level of depression. Thirdly, as expected, stress positively correlated with smartphone addiction, depression and anxiety (r = 0.15, p<0.05; r = 0.92, p<0.01; r = 0.89, p<0.01) respectively. This means that a high level of stress among students invariably yields smartphone addiction and accelerates a high level of anxiety, with depression being worst due to the high level of correlation coefficient (that is, r = 0.92). Hence, the outcome of the results in Table 2 above implies that the second hypothesis, which stated that there would be a significant relationship between smartphone usage and depression, anxiety, and stress among students, was accepted.

Multiple regression analysis was computed for the independent variables in Table 3 above. The first variable, ‘gender’, was not an independent predictor of smartphone addiction, β = -.07; t = -0.94; p>0.05. The same finding was seen with the second independent variable, ‘age’, which was not an independent predictor of smartphone addiction, β = -0.11; t = -1.11; p>0.05. The third independent variable, ‘marital status’, was also not an independent predictor of smartphone addiction, β = 0.08; t = 1.02; p>0.05. The fourth independent variable, ‘religion’, was not an independent predictor of smartphone addiction, β = .07; t = 1.02; p>0.05. The fifth independent variable, ‘level of study’, was also not an independent predictor of smartphone addiction, β = -0.03; t = -.31; p>0.05. However, mental health outcomes comprising of depression, anxiety and stress were independent predictors of smartphone addiction among students, β = -0.11; t = -1.70; p<0.05. Also, significant joint prediction of all predictor variables on smartphone addiction was not recorded F (6, 220) = 1.614; p>0.05. In addition, the independent variables yielded a correlation determination (R) of 0.174 and multiple correlation determination derived from the regression equation (R^2) of 0.030, implying that 3.0% of the variance in smartphone addiction was a result of the merged influence of all independent variables; the adjusted R^2 is 0.004. Therefore, the third hypothesis, which stated that socio-demographic variables and mental health outcomes would jointly predict smartphone addiction among students, was not confirmed.

In **Table 4** above, a moderated multiple hierarchical regression analysis was employed to build a model for predicting student academic performance. At step one, ‘gender’ was entered, and it did contribute significantly to the regression model, F (1, 226) = 4.45, p = 0.036; ‘gender’ explained only 1.9% of the variance in academic performance and was a predictor of academic performance (β = -0.14, SE = 0.14, p = .036). Introducing ‘age’ in the second model F (2, 226) = 2.38, p = 0.05 did change the variance percentage and further led to the prediction of academic performance (β = -0.04, SE = 0.07, p = .045), explaining 2.1 % of the variance in academic performance. This implies that the ‘age’ component alone cannot predict academic performance, but when age merges with the ‘gender’ component, it can make a difference. However, at step three, when ‘marital status was introduced as a moderating variable, the

model equation changed slightly, as a 2.2% variance in academic performance was explained by the three-fold dimension (gender, age and marital status) $F(3, 226) = 1.66$, $p = 0.177$ as against 2.1% and 1.9% in the first two dimensions. Marital status did not significantly moderate the relationship between 'gender' and 'age' ($\beta = -0.04$, $SE = 0.15$, $p = .177$).

At step four, 'religion' was introduced, and the model equation did not improve. Still, it remained at a 2.2% variance contribution of academic performance as explained by gender, age, marital status, and religion, $F(4, 226) = 1.24$, $p = 0.293$. Religion did not significantly moderate the relationship between 'gender,' 'age,' and 'marital status' ($\beta = -0.01$, $SE = 0.22$, $p = .293$). The 'level of study' was introduced in step five. The model equation significantly improved to a 5.2% variance in academic performance explained by gender, age, marital status, and religion, $F(5, 226) = 2.44$, $p = 0.036$. The level of study significantly moderated the relationship between 'gender,' 'age,' 'marital status,' and 'religion' ($\beta = 0.23$, $SE = 0.06$, $p = 0.036$). Lastly, at step six, 'mental health outcomes' was introduced, and the model equation further improved again to a 6.4% variance in academic performance being explained by gender, age, marital status, religion and level of study, $F(6, 226) = 2.52$, $p = 0.022$. Hence, mental health outcomes significantly moderated the relationship between socio-demographic factors ($\beta = 0.11$, $SE = 0.14$, $p = .022$). Thus, the fourth hypothesis, which stated that students' socio-demographic factors and mental health outcomes would significantly influence their academic performance, was confirmed particularly for variables such as 'gender', 'age', 'level of study' and 'mental health outcomes'.

Table 5: Multivariate Analysis of Variance (MANOVA) showing the standardised test scores of socio-demographic factors and mental health outcomes on smartphone addiction and student academic performance.

Model Test	Wilks' Lambda/RLR	*F*	Sig.	R Squared	*Df*	Mean Square
Gender (A)	0.10	0.31	>0.05		1	
Age (B)	0.95	1.38	<0.05		1	
Marital status (C)	0.97	1.35	>0.05		1	
Religion (D)	0.98	0.88	>0.05		1	
Level of study (E)	0.97	0.55	>0.05		1	
MHO (F)	0.98	1.66	>0.05		1	
B * E	0.87	3.41	<0.05		6	
C * F*^{RLR}	0.95	3.69	<0.05		2	
Smart-Phone Addiction (SPA)				0.298		0.24
Academic Performance (AP)				0.419		0.85
Error						
SPA					227	
AP					227	

Note: MHO = Mental Health Outcomes; RLR = Roy's Largest Root

Table 5 revealed the MANOVA overall model tests of significance. A MANOVA has two or more independent and two dependent variables. For some statisticians, the MANOVA compares differences in mean scores between multiple groups and assumes a cause-effect relationship whereby one or more independent, controlled variables (the factors) 'cause' the significant differences in one or more characteristics. Of the six independent variables, a significant main influence on the dependent variables was only recorded for age (Wilks = 0.95, $p<0.05$), while two (2) significant dual interaction effects were reported.

In the first instance, a significant dual interaction effect was found between age and level of study (Wilks = 0.87, $p<0.05$), while the second interaction effect was between marital status and mental health outcomes (Wilks = 0.95, $p<0.05$). In summary, the MANOVA showed that the outcome of some predictor variables predicted smartphone addiction and academic performance among students. Given this, the fifth hypothesis, which stated that those students' socio-demographic factors and mental health outcomes would significantly influence their academic performance and smartphone addiction, was partially retained.

Discussion

The present study investigated mental health outcomes and socio-demographics as determinants of students' academic performance and smartphone addiction at the University of Cape Coast, Ghana. The first hypothesis, which stated that there would be a high prevalence of smartphone addiction among students, was confirmed. The overall 43.6% prevalence of smartphone addiction among the students implies that every 4 in 10 students were addicted to smartphones. And studies have shown that excessive use of smartphones among university students has significant health consequences, including anxiety and depression, leading to poor academic outcomes [2, 14, 42]. Earlier studies have indicated that tertiary education students are the fastest adopters of smartphone technology [19, 3]. It has been reported that university students are addicted to smartphone usage, a phenomenon that takes about 6–8 hours of their time daily [6, 7].

The second hypothesis, which stated that there would be a significant relationship between smartphone usage and depression, anxiety, and stress among students, was accepted. In other words, there was a significant positive relationship between smartphone usage and depression, anxiety, and stress among the students. This finding aligns with a longitudinal study conducted recently in which researchers found a high prevalence of depressive symptoms and stress, especially among students in the first two years of studies. The depression component of mental health outcomes is directly related to smartphone addiction, meaning a high level of depression among students was associated with excessive smartphone usage. To deal with the effects associated with depression, students resort to constant fidgeting with their smartphones [43]. Other studies have also reported the incidence of depression in college students who are addicted to smartphone usage [1, 8, 11, 12, 15, 23, 29].

Additionally, anxiety is directly related to smartphone addiction and depression, respectively. Consequently, it could be argued that a high level of anxiety brings about smartphone addiction and worsens the level of depression due to their close correlations. Furthermore, stress is correlated with smartphone addiction, depression, and anxiety. This implies that a high level of stress among students usually leads to smartphone addiction and invariably stimulates a high level of anxiety, with depression being worst due to the high level of correlation coefficient. Hence, the outcome of the study implies that there is a significant relationship between smartphone usage and depression, anxiety, and stress among students. Mental health outcomes such as depression, anxiety and stress have been associated with smartphone use, which consequently interferes with the daily lives of victims [18, 23, 33, 42].

Furthermore, the third hypothesis, which stated that socio-demographic variables and mental health outcomes would jointly predict smartphone addiction among students, was not confirmed – although mental health outcomes independently predicted smartphone addiction. This finding is in partial agreement with the study conducted in 2024, which explored and evaluated nomophobia and the use of smartphones among university students. The results of the studies showed that nomophobia varied in terms of gender, phone-checking frequency, number of calls made and received per day, as well as number of texts received in a single day predicted nomophobia [4, 9].

The fourth hypothesis, which stated that students' socio-demographic factors and mental health outcomes would significantly influence their academic performance, was confirmed particularly for variables such as 'gender', 'age', 'level of study' and 'mental health outcomes'. This finding is in tandem with a study conducted in 2018, which found an inverse significant association between text messaging and academic performance [21, 38]. Further analysis of the socio-demographic variables suggests that most participants were female (60.8%). This aligns with a study conducted in 2024, which revealed that female participants experienced nomophobia more frequently than male participants. The majority regarding religion were Christians (92.1%) [4]. Also, most students (41%) reported a Cumulative Grade Point Average (CGPA) between 3.0–3.5 and 18.5% with a CGPA of 3.6–4.0. It was also observed that smartphone usage positively and negatively influences students' performance. About 28.2% reported a reduction in CGPA attributed to smartphone usage, and 71.8% reported an increase in CGPA influenced by smartphone applications. The relationship among socio-demographic variables, mental health outcomes and smartphone addiction among university students has been investigated in earlier studies [20, 25, 35, 37].

Lastly, the fifth hypothesis, which stated those students' socio-demographic factors and mental health outcomes would significantly influence their academic performance and smartphone addiction, was partially retained. This finding mirrors the results of a study conducted in 2022, which found that COVID-19 lockdown stress was a result of the joint function of a drug use disorder, neurotic health symptoms, and pathological smartphone use. It was also observed that socio-demographic factors do not independently predict smartphone addiction among students. However, a combination of these factors contributes to smartphone addiction [39]. In a similar study carried out in 2016, although demographic variables such as gender, age, and religion were not independent predictors of smartphone addiction in this study, the results of the survey revealed some association between demographic characteristics of students and smartphone addiction [28], while other studies reported to the contrary [5, 13, 22].

Similarly, socio-demographic factors are not independent predictors of academic performance, but a combination of socio-demographic variables and mental conditions firmly predict academic performance. Therefore, mental health outcomes comprising depression, anxiety and stress were independent predictors of academic performance among students. Findings from the study further revealed that combinations of student's socio-demographic factors such as gender, age, level of study and mental health outcomes significantly influence their academic performance. Limited studies reported the association between students' demographic characteristics, smartphone addiction and academic performance. However, another study conducted in 2016 found a direct relationship between stress, smartphone addiction and academic performance [46]. Also, in a cross-sectional study conducted in 2021 to examine associations between smartphone use, smartphone addiction, and mental health outcomes among college students, it was reported that there was a positive relationship between depression, anxiety and stress, smartphone addiction, and academic performance [27]. Elsewhere, in a study among Lebanese university students, it was found that depression, anxiety, and stress warrant excessive fidgeting with smartphones, which consequently takes away students' time needed to focus on academic activities in school. Findings from the study also pointed to the fact that

students' socio-demographic factors and mental health outcomes would indirectly influence their academic performance and smartphone addiction [33]. In a similar study, demographic characteristics such as age, gender, marital status, religion, period of study and mental health outcomes such as depression, anxiety and stress have been reported in previous studies to influence students' academic performance [1, 36].

Conclusion

Drawing from the results, the study concludes that smartphone addiction is reasonably high among students and should require interventions. The study also observed that depression, anxiety, and stress all had positive correlations with smartphone addiction, which means that the more emotionally distressed students were, the more they tended to indulge in excessive smartphone use. More so, the study noted that mental health outcomes independently predicted smartphone addiction, while academic performance, gender, age and level of study hierarchically predicted academic performance among the study sample. Considering the main and interaction effect of the predictor variables and criterion variables, the study found that only age reported the main impact on the criterion variables (academic performance and smartphone addiction), while on the one hand, age and level of the study reported an interaction effect on the criterion variables. On the other hand, marital status and mental health outcomes showed an interaction effect on the criterion variables.

From the standpoint of policymakers, the findings of this study can stimulate and facilitate shifts in policies toward screening and diagnosis of students to ascertain the level of smartphone addiction while considering their distress levels, performance in school and their individual needs for a more effective positive outcome. Essentially, educationists, social media influencers, information technology professionals, and clinicians can leverage the findings of this study/its importance by creating programs and implementing public policies that will be student-friendly, which in turn can support efforts geared towards academic mentoring, periodic mental health checks, pro bono behavioural modification therapies, extra-mural symposiums and other such related activities which are capable of ensuring an inclusive beneficial future mental health care system for students and young people in higher education.

Limitations and further research

This study adopted standardised assessment tools to assess the mental health outcomes and smartphone addiction among study participants. This means students' subjective views on smartphone addiction were not qualitatively evaluated, which might have limited the students in their quest to share natural experiences on the subject matter. Furthermore, the study focused primarily on the University of Cape Coast nursing students but not the entire student population. Hence, issues may arise in generalisation of findings. Based on this, future studies should incorporate qualitative studies by asking open-ended questions and organising focus group discussions (FGD), which can create a forum for students to give more detailed information. Further studies in the University should endeavour to include students and faculties other than nursing students alone to engage in a broader discussion on mental

health, addiction and the impact on academic performance and make plans to include an intervention or therapeutic phase in the study in an attempt to proffer empirical and psychological solutions to the smartphone addiction problem, which will, in turn, translate to improved academic performance among students.

Abbreviations

SDG-Sustainable Development Goals

Declarations

Ethical approval and consent to participate: This study followed the Helsinki Declaration, and the Institutional Review Board of the University of Cape Coast, Ghana, approved the study protocol. Detailed information was provided, and informed consent was obtained from all participants. All who were willing to participate signed the consent form.

Consent for publication. Not applicable

Data Availability and Material

All data supporting the findings of this study are available from the corresponding author upon reasonable request. However, some restrictions may be raised to protect the rights of the participants in line with ethical considerations.

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Authors' Contributions

JPN: study conceptualisation, theoretical review and discussion of findings; **INI:** Design and Validation of instruments; **DI:** Software analysis and interpretation of results; **IA:** Data processing and analysis; **ITC:** discussion of results; **SOA:** literature review and drafting of the manuscript; **FN:** data gathering and processing **EKA:** Analysis and interpretation. All authors read and made inputs before approving the final manuscript for publication;

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Tables

Tables 3-4 is available in the Supplementary Files section.

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